

Analysis of CPI Film on Bnding Performance of Flexible OLED Panels

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Abstract

Flexible OLED displays have become a development trend. CPI film in a flexible panel can improve the bending performance of the panel. However, CPI film is prone to plastic deformation in the bending process, which will affect the reliability and service life of OLED panel. Finite element analysis is employed in this paper to analyze how CPI film affects the bending performance of a flexible OLED panel. The simulation results show that the optimization of the structure of water-drop-shaped bending can improve the plastic deformation of CPI film. The reasonable position of CPI film in the stacked structure can also improve its plastic deformation. The increase of the thickness of the OCA located below the CPI film can significantly reduce its plastic deformation. The analysis method used in this paper can help improve the bending performance of a flexible OLED panel.

Author Keywords

OLED, flexible panels, CPI, bending performance

1. Introduction

Flexible OLED panels consist of multiple layers of films, and each film is laminated with Optically Clear Adhesive (OCA). Colorless Polyimide (CPI) film has good properties of bending resistance, high light transmission and conductivity. CPI film in a flexible OLED panel can reduce the overall thickness of a panel and improve the optical performance and flexibility. However, CPI film is most prone to plastic deformation in a flexible panel, which can affect the display reliability and service life of flexible panels. Therefore, how to reduce the plastic deformation of CPI film becomes an important issue in designing the stacking structure of flexible OLED panels.

Pengle Dang studied the structural reliability of flexible OLED panels, and the results showed that the stress gradually increases with the decrease of bending radius, and the stress situation becomes more complicated with the decrease of bending radius [1]. Zhengzhou Liu conducted a specific stress and strain analysis of the bending process of flexible OLED panels, and investigated the effects of different stacking structures and backplane stiffness on the panel stress [2]. Niu et al. investigated the effect of bending mode on stress distribution during panel folding, and the experimental results showed that the water-drop-shaped bending can effectively improve the structural stability of flexible panels and reduce the risk of panel damage [3].

In this paper, the finite element simulation method is used to establish the simulation model of flexible OLED panel containing CPI film. The simulation analysis of the flexible module focuses on the strain of the CPI film and the OLED in the flexible module. By adjusting the water-drop-shaped bending structure of the flexible module, the stacking order of the flexible module, and the thickness of each OCA layer in the flexible module, the bending

performance can be improved.

2. Simulation Model of a Flexible OLED Panel

The bending structure of the flexible OLED panel is a kind of water-drop. The simplified water-drop geometry is shown in Figure 1, in which the bending radius is R , the tip of the water-drop is L , and the gap between the two screens is G .

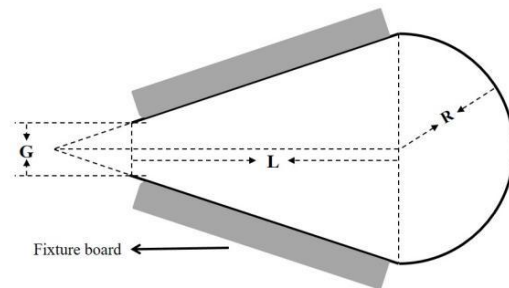


Figure 1. Simplified geometry of water-drop.

The stacked structure of the flexible OLED panel is illustrated in Figure 2, which consists of Cover film, touch panel (TP), polarizer (POL), OCA1 - OCA6, substrate (PET), backplane (BP), CPI, and an OLED light-emitting surface. Some material property parameters are presented in Table 1, in which E and μ are Elastic Modulus and Poisson ratio, respectively. To realize the bending action, the fixture board is necessary, which is also included in Table 1.

Cover film
OCA1
TP
OCA2
POL
OCA3
OLED
OCA4
BP
OCA5
CPI
OCA6
PET

Figure 2. Stacked structure of a flexible OLED panel.

Table 1 Material parameters of each stacked layer in a flexible OLED panel.

Layer	Thickness (um)	E (MPa)	μ
Cover film	56	4000	0.3
OCA1	30	/	/
TP	20	7900	0.3
OCA2	30	/	/
POL	67	22.5	0.3
OCA3	30	/	/
OLED	20	7900	0.3
OCA4	30	/	/
BP	25	3500	0.33
OCA5	30	/	/
CPI	100	2500	0.34
OCA6	30	/	/
PET	75	2500	0.3
Fixture board	80	200000	0.3

OCA is defined with both hyperelasticity and viscoelasticity [4-6], as shown in Table 2 and Table 3.

Table 2 Parameters of the hyperelastic Yeoh model.

C_{10}	C_{20}	C_{30}	D_1	D_2	D_3
0.0675	-0.000253	7.54×10^{-6}	0	0	0

Table 3 Parameters of the viscoelastic Prony model.

i	1	2	3	4	5
g_i	0.591	0.147	0.113	0.064	0.035
t_i	0.019	0.208	1.878	19.196	235.07

A multilinear isotropic reinforcement model was employed for the CPI. The relevant parameters are shown in Table 4.

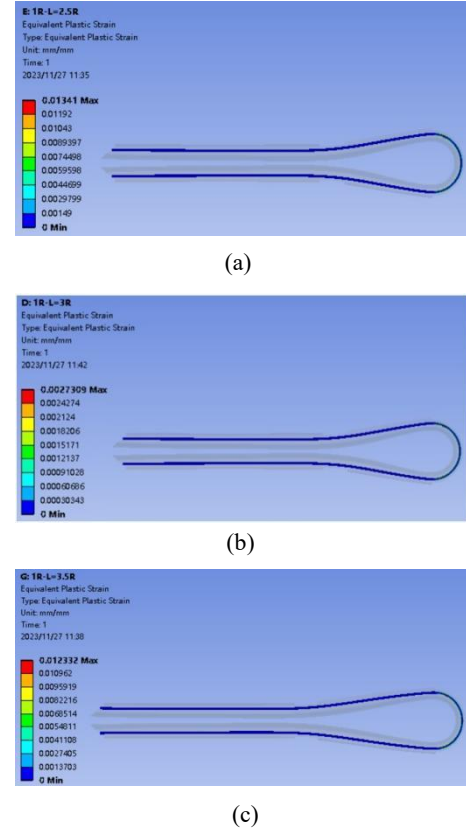
Table 4 Plasticity parameters of CPI film.

Yield Stress/MPa	Plastic Strain Increment
66	0
91	0.02
233	0.71

3. Simulation Analysis and Discussion

3.1 Analysis of the bending structure

The bending radius of a flexible OLED panel is set to 1 mm. The tip of the water drop L is variable. In this paper, three structures of $L=2.5R$, $L=3R$ and $L=3.5R$ are discussed. The simulation results illustrated in Figure 3.

**Figure 3.** Strain simulation distribution of three structures: (a) $L = 2.5R$; (b) $L = 3R$; (c) $L = 3.5R$.

The values of the maximum plastic strain produced on CPI film and the maximum strain produced on the OLED under the three bending structures are obtained respectively. The maximum plastic strain results are shown in Table 5 and the maximum strain results are shown in Table 6.

Table 5 Maximum plastic strain of CPI film under different bending structures.

Maximum Plastic Strain	
$L=2.5R$	0.01341
$L=3.0R$	0.00273
$L=3.5R$	0.01233

Table 6 Maximum strain of the OLED under different bending structures.

Maximum Strain	
$L=2.5R$	0.0099
$L=3.0R$	0.0064
$L=3.5R$	0.0095

The simulation results show that when L is three times of R , the maximum plastic strain of CPI film reaches the minimum, and the maximum strain of the OLED reaches the minimum. It indicates

that by adjusting the structure of the water-drop-shaped bending, the plastic deformation of CPI film can be effectively improved, and the deformation suffered by the OLED is also reduced. When $L=3R$, the maximum plastic strain of CPI film is reduced by 79.6% and the maximum strain of OLED is reduced by 35.4% compared with the case of $L=2.5R$. When $L=3R$, the maximum plastic strain of CPI film is reduced by 77.6% and the maximum strain of the OLED is reduced by 32.6% compared with the case of $L=3.5R$.

3.2 Analysis of the stacked structure

CPI film plays an important role in a flexible panel. CPI film can be used in combination with a backplane and located below the OLED. CPI film can also be used on its own and laminated on top of the OLED. Plastic deformation of CPI film can be reduced by adjusting the stacked structure of the flexible module. Different module stacked schemes are shown in Figure 4.

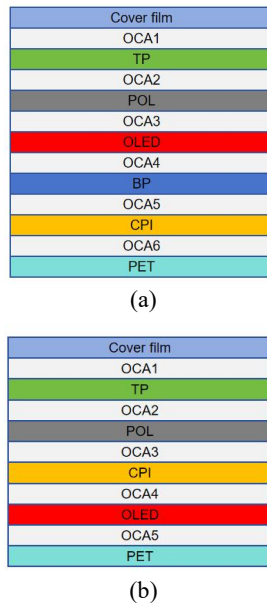


Figure 4. Different stacking schemes for a flexible OLED panel module: (a) Scheme 1; (b) Scheme 2.

The backplane and CPI film are used in Scheme 1, and CPI film is used alone in Scheme 2. The specific simulation results are shown in Figure 5, and the simulation data are shown in Table 7 and Table 8. The simulation results show that the maximum plastic strain on CPI film is reduced by 45.4% and the maximum strain suffered by the OLED is reduced by 3.1% in Scheme 2 compared with Scheme 1. Therefore, changing the stacked structure of the flexible module can also effectively reduce the plastic deformation of CPI film.

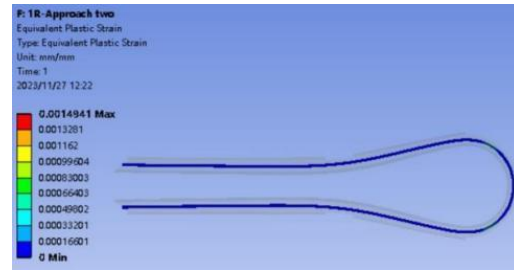
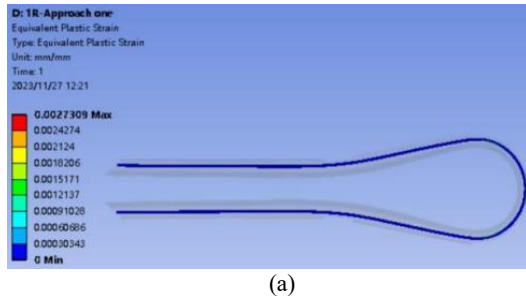


Figure 5. Simulation results for different stacked schemes: (a) Scheme 1; (b) Scheme 2.

Table 7 Maximum plastic strain of CPI film with different stacked schemes.

Maximum Plastic Strain	
BP+CPI	0.00273
CPI	0.00149

Table 8 Maximum strain of the OLED with different stacked schemes.

Maximum Strain	
BP+CPI	0.0064
CPI	0.0062

3.3 Analysis of OCA thickness

The total thickness of OCAs in a flexible module is set to constant, the thicknesses of each OCA are varied. The five thickness combinations, noted as C1-C5, are shown in Table 9.

Table 9 Five different thickness combinations (Thickness unit: μm).

Number of layers	C1	C2	C3	C4	C5
2	17.5	42.5	45	15	30
4	22.5	37.5	30	30	30
6	27.5	32.5	15	45	30
8	32.5	27.5	15	45	30
10	37.5	22.5	30	30	30
12	42.5	17.5	45	15	30

The simulation data are shown in Table 10 and Table 11. The simulation results show that the maximum plastic strain suffered by CPI film decreases significantly with the increase of the thickness of OCA under CPI film. For the OCA thickness combination of C1 and C3, the maximum plastic strain suffered by CPI film was reduced by 61.9% and 79.5%, respectively, compared to C5. In particular, the maximum plastic strain on CPI film reaches the minimum with a plastic strain of 0.00056 with

C3. The maximum strain on the OLED does not change significantly for the five OCA thickness combinations.

Table 10 Maximum plastic strain of CPI film with different OCA thickness combinations.

Maximum Plastic Strain	
C1	0.00104
C2	0.00714
C3	0.00056
C4	0.00887
C5	0.00273

Table 11 Maximum strain of the OLED with different OCA thickness combinations.

Maximum Strain	
C1	0.0068
C2	0.0061
C3	0.0064
C4	0.0061
C5	0.0064

4. Conclusion

In this paper, the finite element analysis model of a flexible OLED panel containing CPI film is established. By changing the bending structure, stacked structure and OCA thickness combination, the maximum plastic strain values of CPI film and the maximum strain values of the OLED in different cases are obtained.

For the bending structure, the maximum plastic strain of the CPI film is minimized when L is three times of the radius, while the maximum strain on the OLED is minimized. For the stacked

structures, the maximum plastic deformation of CPI film and the maximum strain on the OLED can be reduced by changing the stacked order. For different OCA thickness combinations, thicker the thickness of the OCA located below the CPI film, less the maximum plastic strain suffered by CPI film, but the maximum strain suffered by the OLED is not significantly improved. In the case of the thickness combination 3 of OCAs, the thickness of OCA below the CPI film is maximum, and the maximum plastic strain suffered by CPI film reaches the minimum, which effectively improves the plastic performance of CPI film and improves the structural reliability and service life of the flexible OLED panel.

5. Acknowledgments

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6. References

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